QPCode **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION OCTOBER 2026
FIFTH SEMESTER
Discipline Specific Core Course (DSC A) – MG5DSCBTG301- ENZYME TECHNOLOGY
(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

One word answer
Answer any **10** questions.
Each question carries **2** marks

1. What is meant by an enzyme–substrate (ES) complex? [K] / [CO1]
2. Name one enzyme used in the food industry. [U] /[CO4]
3. What is meant by substrate specificity? [K] / [CO1]
4. Define the active site of an enzyme. [K] / [CO1]
5. Why is carrier selection important in enzyme immobilization? [An] / [CO4]
6. What is site-directed mutagenesis? [K] / [CO4]
7. State the role of enzymes in lowering activation energy. [U] / [CO1]
8. Define uncompetitive inhibition. [K] / [CO2]
9. Expand the term abzyme. [U] / [CO4]
10. What is meant by transition state? [K] / [CO1]
11. Define the active site of an enzyme. [K] / [CO1]
12. What is the role of feedback inhibition? [U] / [CO3]

(10X2= 20)

Part B

Short essay
Answer any **5** questions.
Each question carries **4** marks

13. Define enzymes and explain their major characteristics. [U] / [CO1]
14. Describe the applications of enzymes in detergent and textile industries. [App] / [CO4]
15. Differentiate between cofactors, coenzymes, and prosthetic groups. [U] / [CO1]
16. Explain encapsulation as a method of enzyme immobilization. [U] / [CO4]
17. Write a short note on the basic principles of enzyme engineering. [App] / [CO4]
18. What is an allosteric enzyme? [U] / [CO3]

(4X5=20)

Part C

Essay

Answer any **1** question.

Each question carries **10** marks

19. Give an account of enzyme nomenclature and classification. Explain the EC numbering system and the classification of enzymes based on the reactions they catalyse. [U] / [CO1]
20. Explain Michaelis–Menten kinetics. Derive the Michaelis–Menten equation and discuss the significance of Km and Vmax. [U] / [CO2]

(1X10=10)